

I

- 1) a- A Função f é injetiva pois para diferentes valores de x , todas as imagens são diferentes
- b- $f^{-1}([1,2])$

$$y = \frac{x^2}{2} - 2x$$

$$y = \frac{1}{2} - 2 = -\frac{1}{2}$$

$$y = \frac{2^2}{2} - 4 = -2$$

$$c - \text{Mínimo} - x = 2$$

$$\text{Máximo} - x = -1$$

$$d - x = -1$$

$$e - x = 0$$

$$f - |x-2| < \delta \Rightarrow |f(x)+2| < 1$$

$$|f(x)+2| < 1$$

$$\Leftrightarrow \left| \frac{x^2}{2} - 2x + 2 \right| < 1$$

$$\Leftrightarrow -1 < \frac{x^2}{2} - 2x + 2 < 1$$

$$\Leftrightarrow -2 < x^2 - 4x + 4 < 2$$

$$\Leftrightarrow -2 < \underbrace{(x-2)^2} < 2$$

sempre positivo

$$\Leftrightarrow 0 < (x-2)^2 < 2$$

$$\Leftrightarrow 0 < x-2 < 2$$

$$\Leftrightarrow 0 < x < 4$$

II

1) $[-1, +\infty[$

2) $x = -\sqrt{y^2 + 1}$

$\Leftrightarrow 1 - x = \sqrt{y^2 + 1}$

$\Leftrightarrow x^2 = y^2 + 1 \Leftrightarrow y^2 = x^2 - 1 \Leftrightarrow y = -\sqrt{x^2 - 1}$, por' $x \in \mathbb{R}_0^-$

3) $\cos\left(\frac{8\pi}{7}\right) = \cos\left(\pi - \frac{\pi}{7}\right) = -\cos\left(\frac{\pi}{7}\right)$

ângulo cujo $\cos \theta = \frac{\cos 8\pi}{7}$

$\cos \theta = \cos\left(\frac{\pi}{7} - \frac{8\pi}{7}\right)$

$$\sin \theta = \sin \left(-\frac{9\pi}{14} \right)$$

$$\sin \theta = \sin \left(\pi - \frac{5\pi}{14} \right)$$

$$4) f'(u) = (u^2 + 2u + 5)' = 2u + 2$$

$$f'(-2) = 2 \times (-2) + 2 = -2$$

$$f'(a) = 2a + 2$$

$$2a + 2 = \frac{1}{2}$$

$$\text{ou } 4a + 4 = 1$$

$$\text{ou } 4a = -3$$

$$\text{ou } a = -\frac{3}{4}$$

$$\begin{aligned} 5) \left(e^{\sin(u^2+1)} \right)' &= e^{\sin(u^2+1)} \cdot (\sin(u^2+1))' \\ &= \cos(u^2+1) e^{\sin(u^2+1)} \end{aligned}$$

7) f' tem no máximo dois zeros

II

$$\begin{aligned} 1) \quad f \circ f(3) &= f(f(3)) \\ &= f(4) = 5 \end{aligned}$$

$$\begin{aligned} f^{-1} \circ f^{-1}(3) &= f^{-1}(f^{-1}(3)) \\ &= f^{-1}(2) = 1 \end{aligned}$$

$$b - g(x) = (f(x^2))^3$$

$$g'(x) =$$

$$\begin{aligned} 2) \quad a - \lim_{x \rightarrow -\infty} e^x (\sinh x) \\ &= \lim_{x \rightarrow -\infty} e^x \left(\frac{e^{-x} - e^x}{2} \right) \\ &= \lim_{x \rightarrow -\infty} \frac{1 - e^{2x}}{2} \\ &= \frac{1 - e^{-\infty}}{2} = \frac{1 - 0}{2} = \frac{1}{2} \end{aligned}$$

$$b- \lim_{x \rightarrow 0^+} \frac{\sqrt{x} - \sin^2 \sqrt{x}}{x}$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x}}{x} - x \left(\frac{\sin^2 \sqrt{x}}{x^2} \right) \right)$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x}}{x} \right) - \lim_{x \rightarrow 0^+} x \left(\frac{\sin \sqrt{x}}{x} \right)^2$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x}}{x} \right) - \lim_{x \rightarrow 0^+} x \times \lim_{x \rightarrow 0^+} \left(\frac{\sin \sqrt{x}}{x} \right)$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x}}{x} \right) - \lim_{x \rightarrow 0^+} x \times \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x} \times \frac{\sin \sqrt{x}}{x \times \sqrt{x}}}{1} \right)$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x}}{x} \right) - \lim_{x \rightarrow 0^+} \left(x \times \frac{\sqrt{x}}{x} \right) \times \lim_{x \rightarrow 0^+} \frac{\sin \sqrt{x}}{\sqrt{x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{x} - 0 \times 1$$

$$> \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sqrt{x} \times \sqrt{x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}}$$

$$= \frac{1}{0^+} = +\infty$$

3)

$$(x-2)^2 + y^2 = 4$$

$$\Leftrightarrow y^2 = 4 - (x-2)^2$$

$$\Leftrightarrow y = \pm \sqrt{4 - (x-2)^2}$$

$$\Leftrightarrow y = -\sqrt{4 - (x-2)^2} \quad \text{pois } y < 0$$

$$f(2) = -\sqrt{4 - (2-2)^2} = -2$$

$(2, -2)$ é o mínimo a função

Daremos que:

$$|f(x) + 2| < 1/2$$

$$| -\sqrt{4-(x-2)^2} + 2 | < 1/2$$

$\underbrace{\quad\quad\quad}$
 Sempre maior que 0 pois o mínimo é igual
 $x-2$ logo este || tem como valor
 mínimo 0

$$-\sqrt{4-(x-2)^2} + 2 < 1/2$$

$$\Leftrightarrow -\sqrt{4-(x-2)^2} < -3/2$$

$$\Leftrightarrow \sqrt{4-(x-2)^2} > 3/2$$

$$\Leftrightarrow 4 - (x-2)^2 > 9/4$$

$$\Leftrightarrow (x-2)^2 < -(-\frac{7}{4})$$

$$\Leftrightarrow (x-2)^2 < \frac{7}{4}$$

$$\Leftrightarrow x-2 < \sqrt{\frac{7}{4}}$$

$$\Leftrightarrow x-2 < \frac{\sqrt{7}}{2}$$

$$\frac{9}{4} - 4 =$$

$$3) \quad \frac{1}{10^{n+1}} = 0, \underbrace{00 \dots 01}_{n+1 \text{ zeros}}$$

$$\frac{1}{10^n} = 0, \underbrace{00 \dots 01}_n$$